

Fast Radar Pulse Deinterleaving via Multi-Head Attention Broad Learning System

Longkun Kuang

*College of Electronic Engineering
National University of Defense
Technology
Hefei, China
kuanglongkun20@nudt.edu.cn*

Mingyang Du

*College of Electronic Engineering
National University of Defense
Technology
Hefei, China
duminyang17@nudt.edu.cn*

Zhihui Li

*College of Electronic Engineering
National University of Defense
Technology
Hefei, China
lizhihui_16@163.com*

Junpeng Shi

*College of Electronic Engineering
National University of Defense
Technology
Hefei, China
shijunpeng20@nudt.edu.cn*

Xuan Chen

*College of Electronic Engineering
National University of Defense
Technology
Hefei, China
chenxuan24@nudt.edu.cn*

Abstract—In modern electronic reconnaissance, the proliferation of radars employing complex Pulse Repetition Interval (PRI) modulation presents significant challenges to signal deinterleaving. Conventional algorithms often grapple with performance degradation in complex electromagnetic environments, while emerging deep learning approaches are hindered by high computational latency and a reliance on extensive labeled datasets. To address these limitations, this paper proposes a Multi-Head Attention Broad Learning System (MHA-BLS). This framework eschews iterative backpropagation, integrating attention-based feature extraction within a broad learning architecture. By employing random orthogonal initialization, the proposed method ensures feature diversity while maintaining system stability. Numerical simulations demonstrate the effectiveness and computational efficiency of MHA-BLS in small-sample scenarios.

Index Terms—radar signal deinterleaving, broad learning system, multi-head attention, small-sample learning.

I. INTRODUCTION

Electronic Support Measures (ESM) systems are pivotal for situational awareness in modern electronic warfare, tasked with sensing the electromagnetic spectrum to intercept and analyze hostile radar emissions [1]. As the initial stage of the ESM processing chain [2], radar signal sorting (RSS), also referred to as deinterleaving, aims to segregate interleaved pulse streams into sequences associated with specific emitters. Effective deinterleaving is imperative for subsequent tasks such as emitter classification, identification, and threat assessment.

Existing deinterleaving methodologies generally fall into two categories: TOA-based and multi-parameter-based approaches [3]. Algorithms that utilize pulse description word (PDW)—specifically parameters like Time of Arrival (TOA), Radio Frequency (RF), Direction of arrival (DOA), and Pulse

Width (PW)—have historically been effective against conventional radars. Representative techniques include the Cumulative Difference Histogram (CDIF) [4], sequential difference histogram (SDIF) [5], and PRI Transform [6]. However, the operational landscape has shifted with the deployment of Low Probability of Intercept (LPI) radars and multifunction radars (MFRs) [7]. These systems utilize agile parameters and complex modulation strategies, such as jittered, staggered, and dwell-and-switch PRI, rendering the parameter distributions highly dynamic and overlapping. Consequently, traditional histogram and clustering methods frequently suffer from severe performance degradation [8].

To circumvent these limitations, recent research has pivoted towards deep learning (DL) techniques. Architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have been adapted for signal sorting, demonstrating enhanced adaptability to complex modulations [9]–[11]. Despite their efficacy, DL-based methods rely on iterative backpropagation results in high computational cost and latency, and their performance strongly depends on large-scale labeled datasets, which are often unavailable in non-cooperative ESM scenarios. Training deep networks on such limited datasets typically leads to overfitting and poor generalization. The Broad Learning System (BLS) offers a flat alternative that replaces iterative backpropagation with analytical pseudoinverse solutions, thereby achieving extremely rapid training speeds while maintaining robust generalization [12].

The standard BLS relies on random projections for feature extraction, which inherently constrains its feature representation capability. The attention mechanism is widely used to capture global dependencies and enhance representation capacity in complex sequence modeling tasks [13]. To bridge this

gap while preserving computational efficiency, we introduce a Multi-Head Attention Broad Learning System (MHA-BLS) tailored for radar signal deinterleaving under small-sample constraints. We incorporate a MHA mechanism to refine feature extraction and employ random orthogonal initialization to bolster model stability. This integration fortifies the model's capability to disentangle complex interleaved pulse streams, ensuring robust sorting accuracy while maintaining the rapid training efficiency inherent to the broad learning architecture.

The remainder of this paper is structured as follows: Section II outlines the problem formulation. Section III details the MHA-BLS architecture. Section IV presents numerical experiments, followed by the conclusion in Section V.

II. PROBLEM FORMULATION

A. PDW Data Representation

We consider a non-cooperative electronic reconnaissance scenario where a receiver intercepts a dense, interleaved stream of radar pulses from M distinct emitters. The receiver processes the radio frequency input to generate a sequence of Pulse Description Words (PDWs).

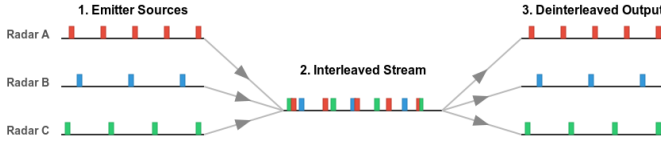


Fig. 1. Illustration of radar signal deinterleaving.

Let the intercepted sequence be denoted as $\mathbf{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N\}$, where N represents the total pulse count. Each pulse is characterized by a feature vector $\mathbf{p}_n \in \mathbb{R}^D$ comprising measured parameters:

$$\mathbf{p}_n = [TOA_n, RF_n, PW_n, DOA_n]^T, \quad (1)$$

where TOA , RF , PW , and DOA denote Time of Arrival, Radio Frequency, Pulse Width, and Direction of Arrival, respectively.

The deinterleaving task is formulated as learning a mapping from the pulse sequence $\mathbf{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N\}$ to a label sequence $\mathbf{Y} = \{y_1, y_2, \dots, y_N\}$, where y_i identifies the specific emitter source of pulse $\mathbf{p}_i$.

B. PRI and Modulation Patterns

The Pulse Repetition Interval (PRI) is defined as the interval between consecutive pulses:

$$PRI_i = TOA_{i+1} - TOA_i. \quad (2)$$

Modern radars employing agile modulations introduce ambiguity into PRI-based sorting. Common modulation types include:

- 1) **Jitter PRI**: The PRI varies randomly around a center value.
- 2) **Stagger PRI**: The PRI cycles through a predefined sequence of several distinct values.
- 3) **Dwell&Switch PRI**: The PRI remains constant for a burst of pulses before switching to another value.

C. Input Preprocessing

To enhance signal stationarity, we utilize the first-order difference of the TOA (DTOA) rather than absolute TOA values. Given the disparate scales of PDW parameters (e.g., GHz for RF vs. μs for PW), we apply min-max normalization to standardize the input:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}. \quad (3)$$

III. MHA-BLS FRAMEWORK

To tackle the dual challenges of small-sample availability and real-time processing requirements, we propose the Multi-Head Attention Broad Learning System (MHA-BLS). Unlike deep learning models that depend on deep, layer-wise back-propagation, MHA-BLS expands features horizontally and solves for output weights analytically. Additionally, we introduce a Random Orthogonal Initialization strategy to mitigate the instability often observed in random mapping networks [14]. The detailed structure is depicted in Fig. 2.

A. Random Orthogonal Weight Initialization

In standard BLS and random vector functional link networks, weights are typically drawn from a standard normal distribution, $\mathbf{W} \sim N(0, 1)$. While computationally efficient, this can result in linear dependence among feature nodes and an ill-conditioned system matrix, particularly in high-dimensional spaces.

To alleviate this, we enforce orthogonality in the projection matrices. Let $\mathbf{W} \in \mathbb{R}^{d_{in} \times d_{out}}$ be the target weight matrix. We generate it via the QR decomposition of a random Gaussian matrix $\mathbf{G}$:

$$\mathbf{G} \sim N(0, 1)^{d_{in} \times d_{out}}, \quad \mathbf{G} = \mathbf{Q}\mathbf{R}, \quad (4)$$

where d denotes the matrix dimensionality, and $\mathbf{R}$ is an upper triangular matrix and $\mathbf{Q}$ satisfies $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$. We then assign $\mathbf{W} = \mathbf{Q}$.

This orthogonal initialization preserves the norm of error gradients and ensures that the attention heads capture diverse subspaces of the PDW data, preventing "mode collapse" where multiple heads learn redundant features.

B. Multi-Head Attention-Based Feature Mapping

We integrate a multi-head attention mechanism to replace the standard feature mapping layer of BLS. MHA captures adaptive cross-pulse feature interactions that a fixed random projection cannot express.

Let the normalized PDW input be $\mathbf{X} \in \mathbb{R}^{N \times D}$. The input serves simultaneously as the Query ($\mathbf{Q}_{in}$), Key ($\mathbf{K}_{in}$), and Value ($\mathbf{V}_{in}$). The system employs n parallel attention heads to capture information from different representation subspaces. The output of the $\mathbf{Head}_i$ is computed as:

$$\mathbf{Head}_i = \text{softmax} \left(\frac{(\mathbf{X}\mathbf{W}_i^Q)(\mathbf{X}\mathbf{W}_i^K)^T}{\sqrt{d_k}} \right) (\mathbf{X}\mathbf{W}_i^V), \quad (5)$$

where $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V \in \mathbb{R}^{D \times d_k}$ are the orthogonal projection matrices for the $\mathbf{Head}_i$, and d_k is the dimensionality of

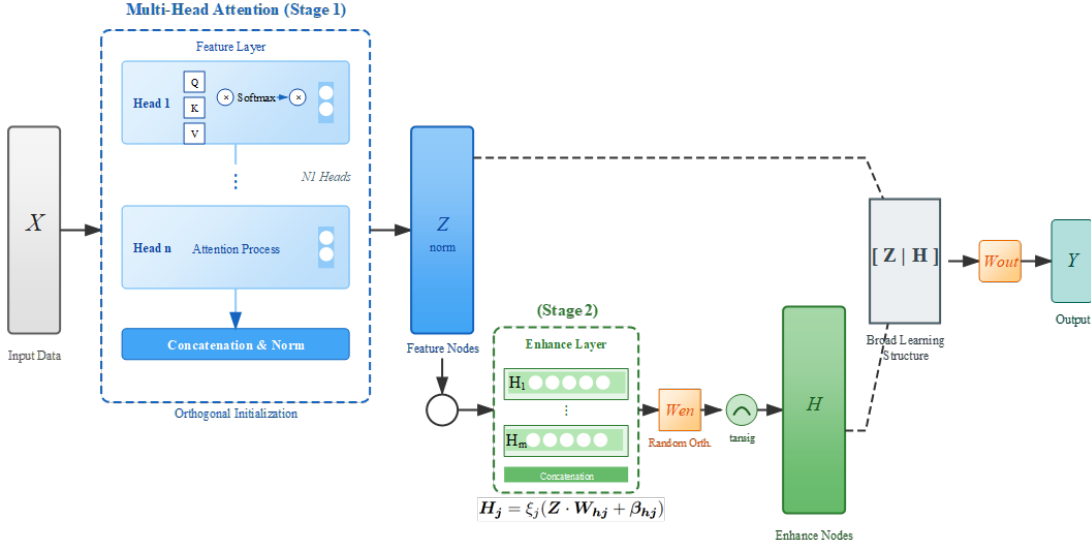


Fig. 2. The model architecture of the proposed MHA-BLS.

the key vectors. The scaling factor $\sqrt{d_k}$ is employed to prevent vanishing gradients in the softmax function.

The outputs from all h heads are concatenated to form the mapped feature nodes $Z \in \mathbb{R}^{N \times (h \cdot d_k)}$:

$$Z = [Head_1, Head_2 \cdots Head_n]. \quad (6)$$

This matrix Z replaces the traditional mapped features in BLS, providing a richer, context-aware representation of the signal environment.

The MHA layer complexity is $\mathcal{O}(N^2d + Nd^2)$, dominated by the pairwise attention term when $N \gg d$.

C. Enhancement Layer Feature Mapping

To enrich the representation, the mapped features Z are further expanded into a broad "enhancement layer." This layer introduces nonlinearity and increases the effective width of the network. m groups of enhancement nodes are generated via random projection and nonlinear activation:

$$H_j = \xi_j(Z \cdot W_{hj} + \beta_{hj}), \quad (7)$$

where $\xi_j(\cdot)$ is the activation function (e.g., $\tanh$), and W_{hj} , β_{hj} are the weights and biases for the j -th enhancement group. The aggregated enhancement nodes are:

$$H = [H_1, H_2 \cdots H_m]. \quad (8)$$

The feature nodes generated by the feature mapping layer and the enhancement nodes from the enhancement layer are concatenated to form the complete input to the output layer, denoted as A . The matrix A is given by

$$A = [Z, H]. \quad (9)$$

The final output is determined by the linear projection:

$$Y = AW^{\text{out}}. \quad (10)$$

where Y denotes the output layer, and W^{out} connects the broad structure to the label space. Unlike deep learning models that require iterative gradient descent to update all weights, BLS treats the optimization of W^{out} as a ridge regression problem. The objective function is formulated as:

$$\arg \min_{W^{\text{out}}} \|AW^{\text{out}} - Y\|_2^2 + \lambda \|W^{\text{out}}\|_2^2. \quad (11)$$

where λ is the regularization coefficient to control model complexity and prevent overfitting on small samples.

Consequently, the optimal output weights W^{out} are derived analytically via ridge regression:

$$W^{\text{out}} = (A^T A + \lambda I)^{-1} A^T Y, \quad (12)$$

This closed-form solution allows the entire network to be trained in a single step, resulting in the orders-of-magnitude speedup observed in our experiments.

IV. NUMERICAL EXPERIMENTS

We simulated a complex electromagnetic environment comprising four distinct radar emitters. As detailed in Table I, the parameters include complex modulations (Jitter, Stagger, Dwell&Switch) to mimic modern LPI radars. The dataset consists of 8,000 pulses interleaved chronologically. We compared MHA-BLS against LSTM, temporal convolutional network (TCN) [10], K-means, and standard BLS. The dataset was split 8:2 for training and testing.

The MHA-BLS configuration included 32 heads with 24 feature nodes each, 800 enhancement nodes, a scaling factor $s = 0.8$, and regularization $\lambda = 0.0001$. Comparative DL models utilized a learning rate of 0.0001 over 200 epochs with a batch size of 32. In the K-means algorithm, the number of clusters K is set to 4, and the cluster-to-class mapping is performed via majority voting.

In order to comprehensively evaluate the sorting performance, the Precision, recall and F1-score are adopted as the

TABLE I
PARAMETERS OF SIMULATED RADAR EMITTERS

ID	PRI Mod.	PRI/ μ s	PW/ μ s	RF/GHz	DOA/ $^\circ$
1	Fixed	520	10–30	8.1–8.4	34–37
2	Jitter	350 ($J = 0.25$)	30–50	8.15–8.3	36–39
3	Stagger	100/280/800	20–40	8.35–8.6	33–37
4	Dwell&Switch	400×15	25–45	8.25–8.5	36.5–38
		600×20			
		900×10			

primary evaluation metric in the comparative experiments. Precision, recall and F1-score are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (13)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (14)$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (15)$$

where TP denotes the number of true positives, FP denotes the number of false positives, and FN denotes the number of false negatives. Precision, Recall, and F1-score are computed for each radar emitter class independently and then macro-averaged across all classes.

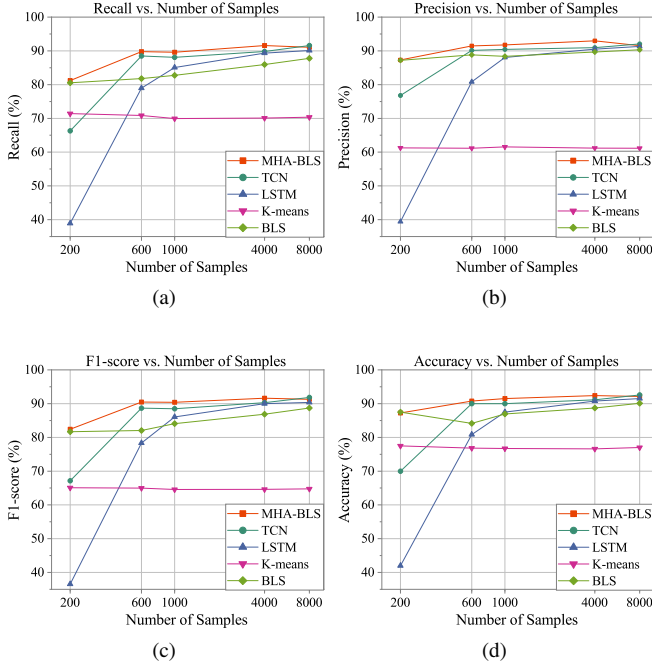


Fig. 3. Deinterleaving performance vs. number of training samples with (a) Recall, (b) Precision, (c) F1-score, and (d) Accuracy. The x-axis is on a $\log_{10}$ scale.

Fig. 3 illustrates the comparative results across four performance indicators as a function of sample size. In small-sample regimes (fewer than 1,000 pulses), MHA-BLS exhibits a distinct advantage across all metrics, consistently outperforming LSTM, TCN, and K-means. This superiority is quantitatively corroborated by Table II: with merely 200 pulses, MHA-BLS

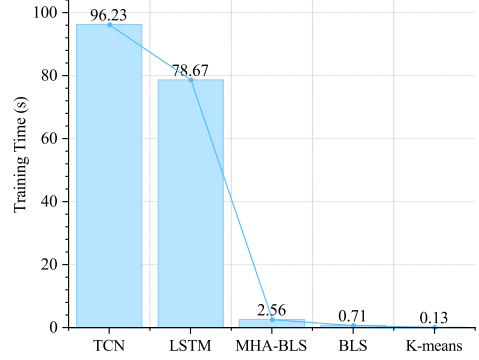


Fig. 4. Comparison of training time with 8,000 samples.

TABLE II
PERFORMANCE OF DIFFERENT ALGORITHMS WITH 200 SAMPLES.

Algorithm	F1-score (%)	Recall (%)	Precision (%)
MHA-BLS	82.42	81.22	87.32
M'HA-BLS ^a	81.84	80.56	86.82
TCN	67.14	66.29	76.77
LSTM	36.53	38.90	39.42
K-means	65.09	71.43	61.26

^a Ablation variant with non-orthogonal random weight.

achieves an F1-score of 82.42%, significantly surpassing the baselines. The comparison with M'HA-BLS further verifies the contribution of orthogonal initialization, which consistently improves all three metrics and yields more stable performance. While TCN marginally overtakes MHA-BLS as the dataset expands to 8,000 samples, this gain comes at the expense of substantial computational overhead due to iterative back-propagation. In contrast, MHA-BLS retains a superior balance between accuracy and efficiency, leveraging its closed-form solution to avoid the prolonged training latency typical of deep models.

Fig. 4 highlights this efficiency gap. Training TCN and LSTM requires approximately 96.23 seconds and 78.67 seconds with 8000 samples, respectively. In stark contrast, MHA-BLS completes training in just 2.56 seconds. This orders-of-magnitude reduction in latency, combined with robust accuracy in small data conditions, positions MHA-BLS as a superior candidate for agile cognitive electronic warfare systems.

V. CONCLUSIONS

This paper presented a Multi-Head Attention Broad Learning System (MHA-BLS) for efficient radar signal deinterleaving. By integrating attention-based feature extraction with an analytically solved broad learning architecture, the proposed method effectively circumvents the high latency and data dependency issues inherent in deep learning models. Numerical experiments confirm that MHA-BLS offers a superior trade-off between accuracy and speed, particularly in small-sample scenarios typical of non-cooperative reconnaissance. Future work will focus on further expanding the topological structure of MHA-BLS to enhance its representational capacity.

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